Description

Illustration





Figure caption

1. High density concrete used is nuclear generation plants as a radiation shield. 2. The concrete is used at the base of offshore drilling platforms to improve stability. Images used under license from https://stock.adobe.com/

The material

High density concrete, also called high weight concrete is concrete with density about 50% higher than the standard Portland cement concrete. High density is achieved by using high density aggregates such as barite, steel shot, ferrophosphorus hematite or other minerals. High density concrete is mainly used for radiation shielding (X-rays, gamma rays and neutrons) and for counterweights. It is an important material for the nuclear industry. Density varies significantly depending on the aggregate used.
Note that the values quoted are for concrete with Baryte and steel shot aggregate.

Compositional summary

0.58:1:4.6:6.4 Water:OPC:Fine:Coarse (Aggregate=Baryte or Steel Shot)

General properties

Density	4.9e3	-	5.5e3	kg/m^3
Price	* 0.189	-	0.234	GBP/kg

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✗
Services	✗

Mechanical properties

Young's modulus	*	40.2	-	41.6	GPa
Shear modulus	*	16.5	-	17	GPa
Bulk modulus	*	23.9	-	24.8	GPa
Bending modulus		40.2	-	41.6	GPa
Poisson's ratio		0.2	-	0.24	
Yield strength (elastic limit)	*	3.1	-	3.7	MPa
Tensile strength	*	3.1	-	3.7	MPa
Compressive strength	*	30.6	-	36.6	MPa
Bending strength	*	3.7	-	4.4	MPa
Elongation		0.01			% strain
Hardness - Vickers	*	9.5	-	10.5	HV
Fatigue strength at 10^7 cycles	*	1.52	-	2.38	MPa
Fracture toughness		0.35	-	0.45	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.01	-	0.02	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		0.26	-	0.435	m.°C/W
Thermal expansion coefficient		10	-	12	µstrain/°C
Specific heat capacity	*	473	-	867	J/kg.°C
Melting point		930	-	1.2e3	°C
Maximum service temperature	*	627	-	857	°C
Minimum service temperature		-160	-	-150	°C
Flammability		Non-flammable			
Emissivity		0.63	-	0.97	

Hygro-thermal properties

Water vapor permeability		2e-12	-	6.01e-12	kg.m/s.m^2.Pa
Frost resistance		Average			

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity	*	2e12	-	2e13	µohm.cm
Dielectric constant (relative permittivity)	*	8	-	12	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.01	
Dielectric strength (dielectric breakdown)	*	0.5	-	2	MV/m

Optical properties

Natural color

Gray

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No			

Acoustic properties

Sound absorption	Poor			
Sound isolation	Good			

Processability (scale 1 = impractical to 5 = excellent)

Castability	4	-	5	
Moldability	3	-	4	

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Unacceptable
Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.113	-	0.125	kg/kg	
Embodied energy (virgin grade production)	* 0.923	-	1.02	MJ/kg	
Water Usage	* 2.77	-	3.06	l/kg	

Material processing: energy

Machining, grinding, embodied energy	* 16.4	-	18.2	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.07	-	1.2	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.0113	-	0.0125	kg/kg
Embodied energy, recycling	* 0.129	-	0.143	MJ/kg
Recycle fraction in current supply	14			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Links

Reference